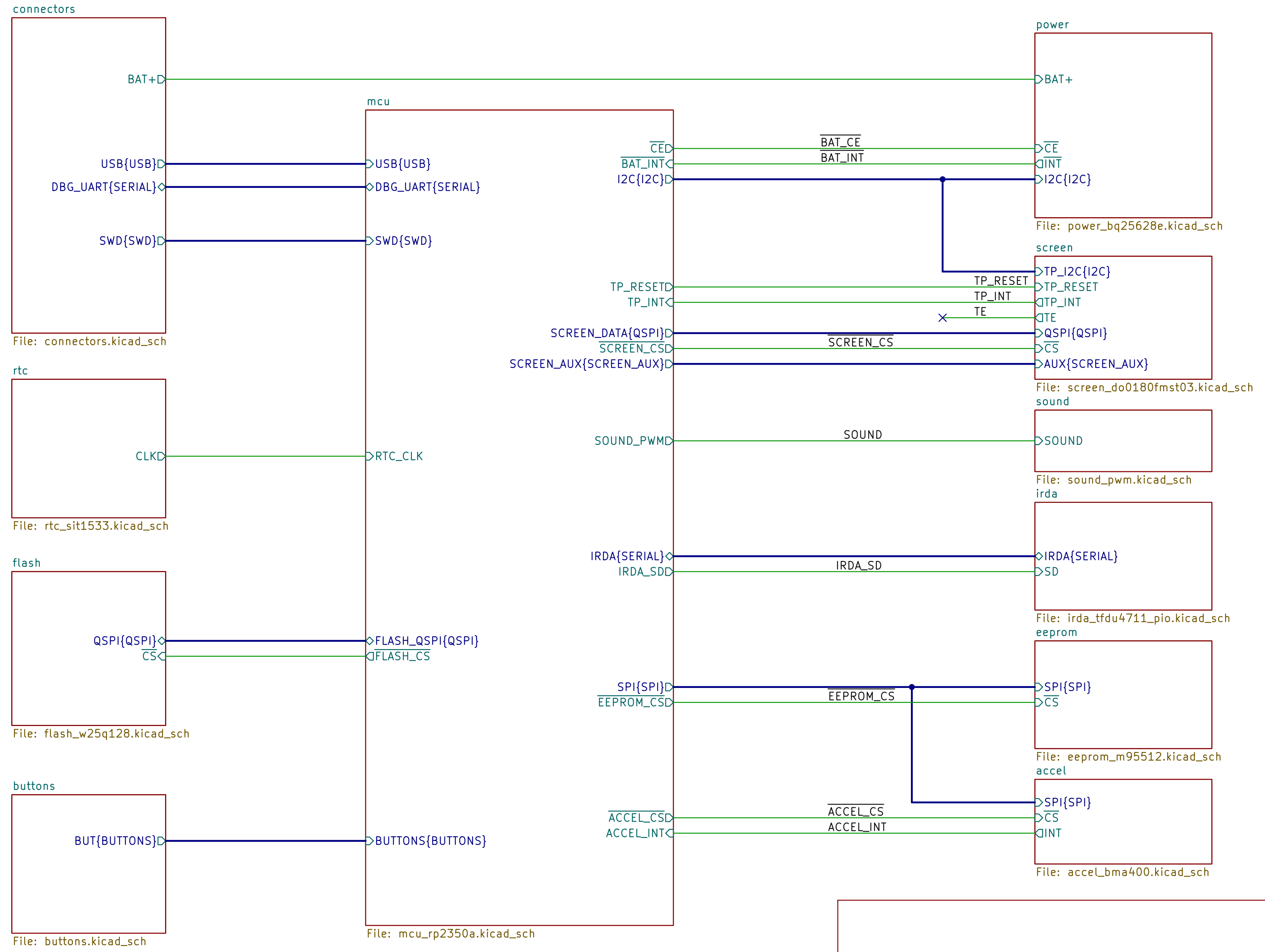
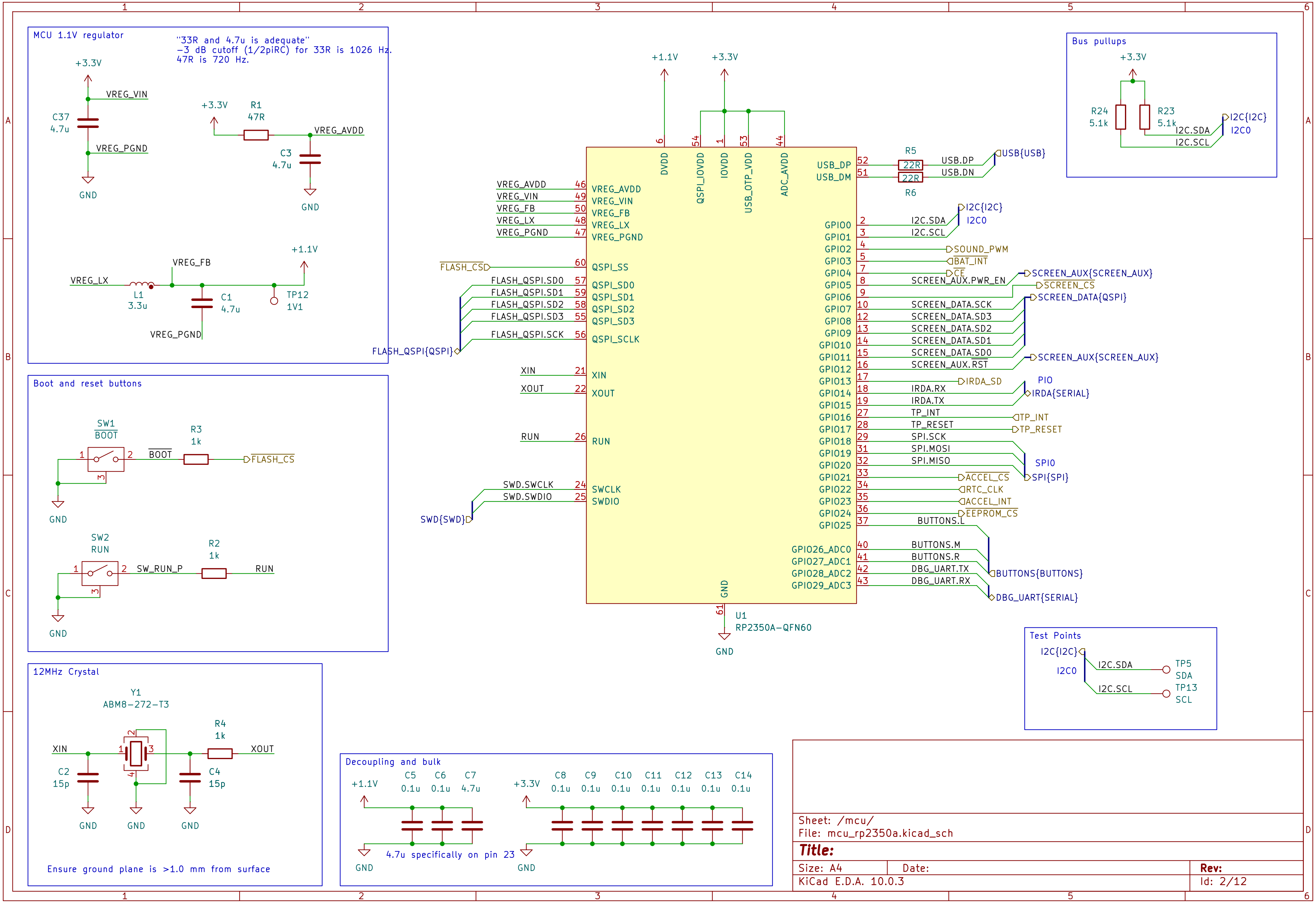
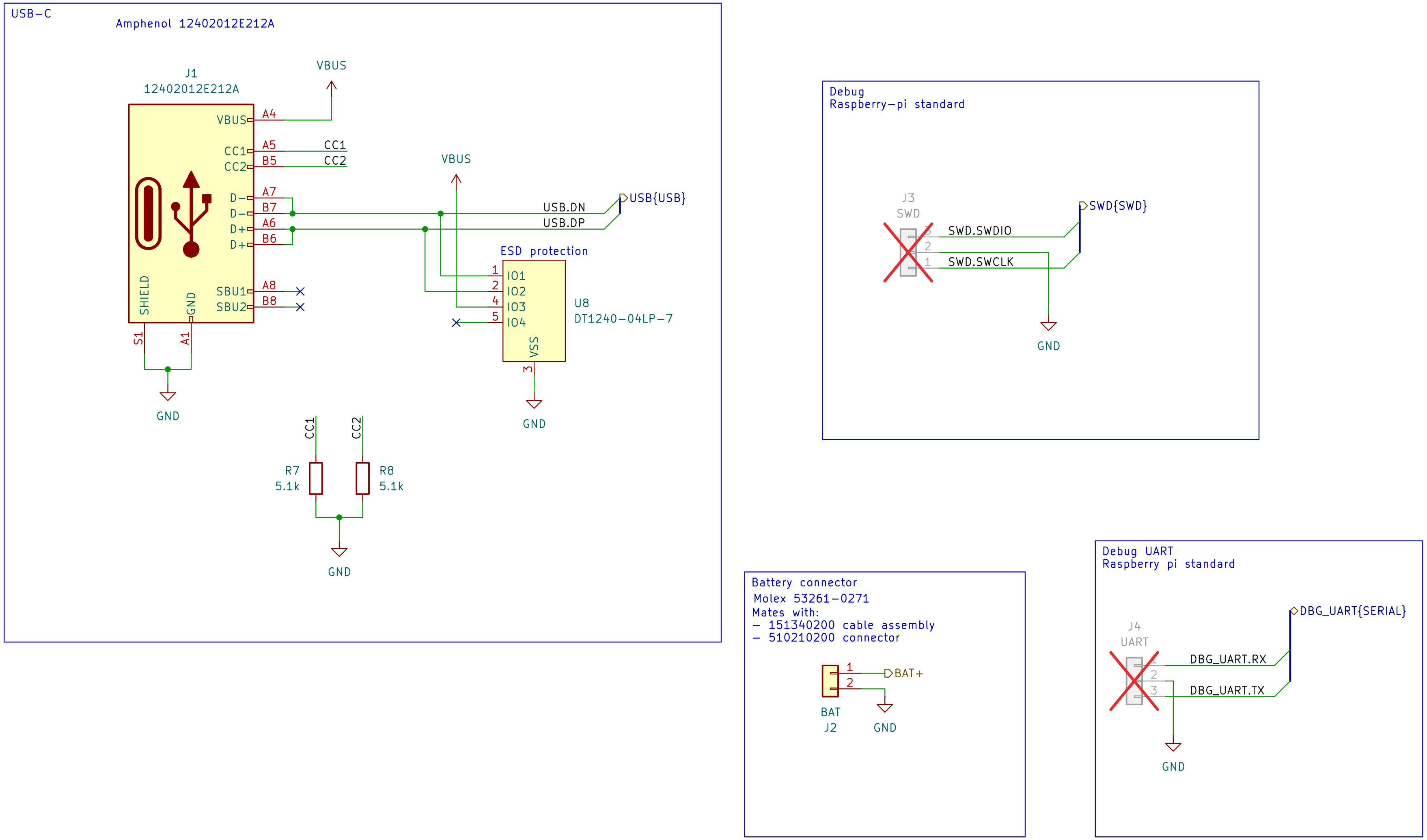


Sheet: /
File: picowalker-v0.4.kicad_sch

Title: picowalker

Size: A4	Date:	**Rev: 0.4.1**
KiCad E.D.A. 10.0.3		Id: 1/12

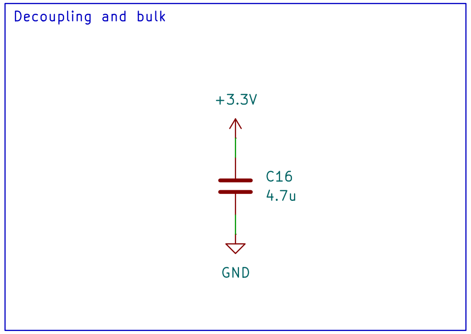

MCU 1.1V regulator
"33R and 4.7u is adequate"
-3 dB cutoff (1/2piRC) for 33R is 1026 Hz.
47R is 720 Hz.
+3.3V
VREG_VIN
C37
4.7u
VREG_PGND
GND
R1
47R
VREG_AVDD
C3
4.7u
VREG_FB
+1.1V
VREG_LX
L1
3.3u
C1
4.7u
TP12
1V1
Boot and reset buttons
SW1
BOOT
R3
1k
FLASH_CS
SW2
RUN
SW_RUN_P
R2
1k
RUN
12MHz Crystal
Y1
ABM8-272-T3
R4
1k
XIN
XOUT
C2
15p
C4
15p
Ensure ground plane is >1.0 mm from surface
Decoupling and bulk
C5 C6 C7
0.1u 0.1u 4.7u
C8 C9 C10 C11 C12 C13 C14
0.1u 0.1u 0.1u 0.1u 0.1u 0.1u 0.1u
4.7u specifically on pin 23
DVDD
QSPI_IOVDD
IOVDD
USB_OTP_VDD
ADC_AVDD
VREG_AVDD
VREG_VIN
VREG_FB
VREG_LX
VREG_PGND
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
XIN
XOUT
RUN
SWCLK
SWDIO
FLASH_QSPI.SD0
FLASH_QSPI.SD1
FLASH_QSPI.SD2
FLASH_QSPI.SD3
FLASH_QSPI.SCK
FLASH_QSPI{QSPI}
SWD.SWCLK
SWD.SWDIO
SWD{SWD}
USB_DP
USB_DM
R5
22R
R6
22R
USB.DP
USB.DN
USB{USB}
GPIO0 … GPIO25
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
I2C.SDA
I2C.SCL
I2C{I2C}
I2C0
SOUND_PWM
BAT_INT
CE
SCREEN_AUX.PWR_EN
SCREEN_AUX{SCREEN_AUX}
SCREEN_CS
SCREEN_DATA.SCK
SCREEN_DATA.SD3
SCREEN_DATA.SD2
SCREEN_DATA.SD1
SCREEN_DATA.SD0
SCREEN_DATA{QSPI}
SCREEN_AUX.RST
IRDA_SD
IRDA.RX
IRDA.TX
PIO
IRDA{SERIAL}
TP_INT
TP_RESET
SPI.SCK
SPI.MOSI
SPI.MISO
SPI0
SPI{SPI}
ACCEL_CS
RTC_CLK
ACCEL_INT
EEPROM_CS
BUTTONS.L
BUTTONS.M
BUTTONS.R
BUTTONS{BUTTONS}
DBG_UART.TX
DBG_UART.RX
DBG_UART{SERIAL}
U1
RP2350A-QFN60
Bus pullups
R24
5.1k
R23
5.1k
Test Points
TP5
SDA
TP13
SCL
Sheet: /mcu/
File: mcu_rp2350a.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 2/12

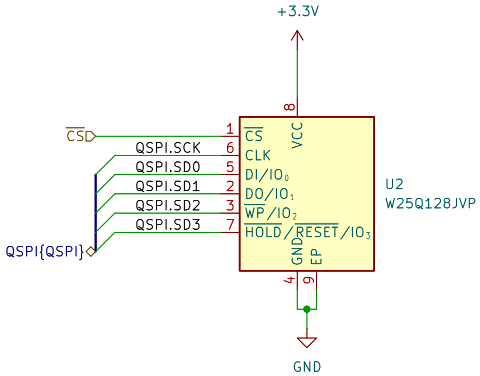


Sheet: /connectors/
File: connectors.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 3/12





Sheet: /flash/
File: flash_w25q128.kicad_sch

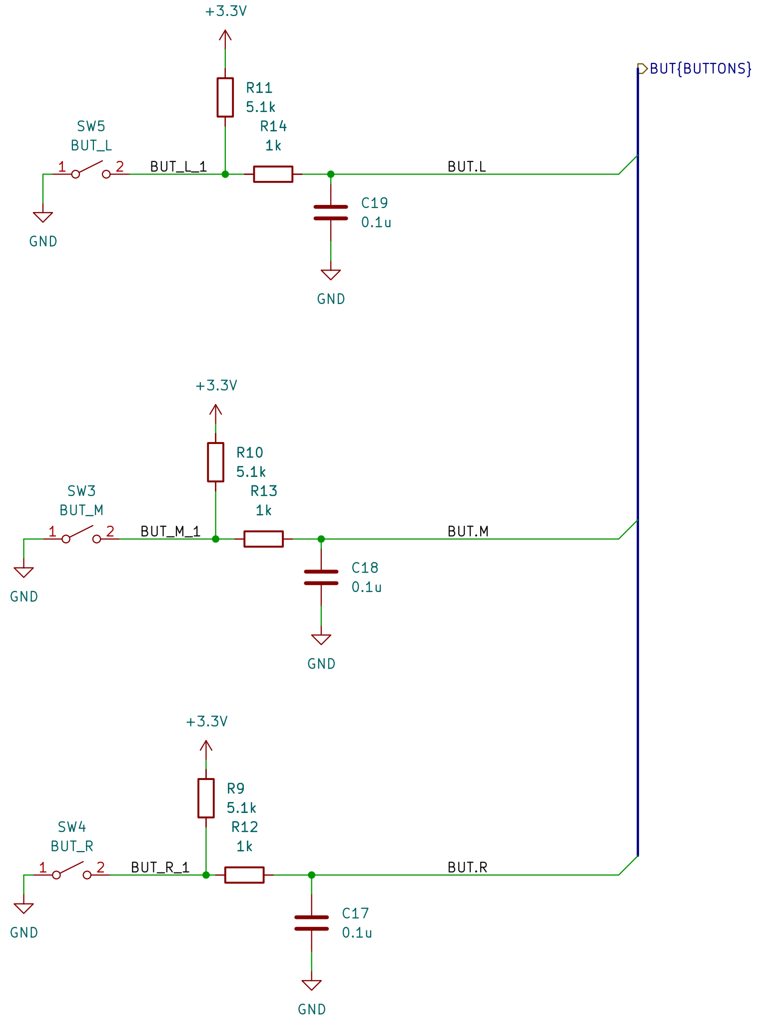
Title:

Size: A4 Date:

KiCad E.D.A. 10.0.3

Rev:

Id: 4/12

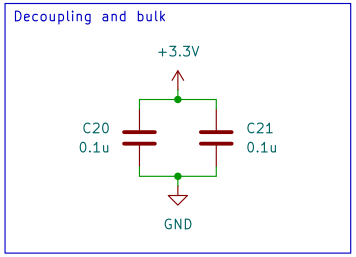


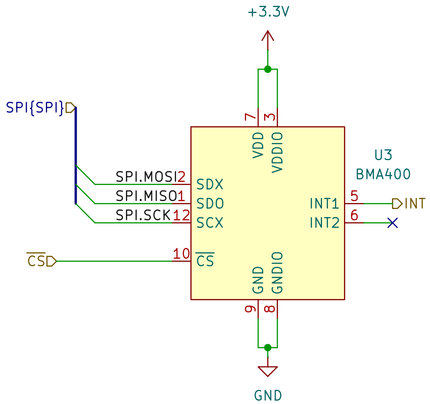
Sheet: /buttons/
File: buttons.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.3 | Id: 5/12

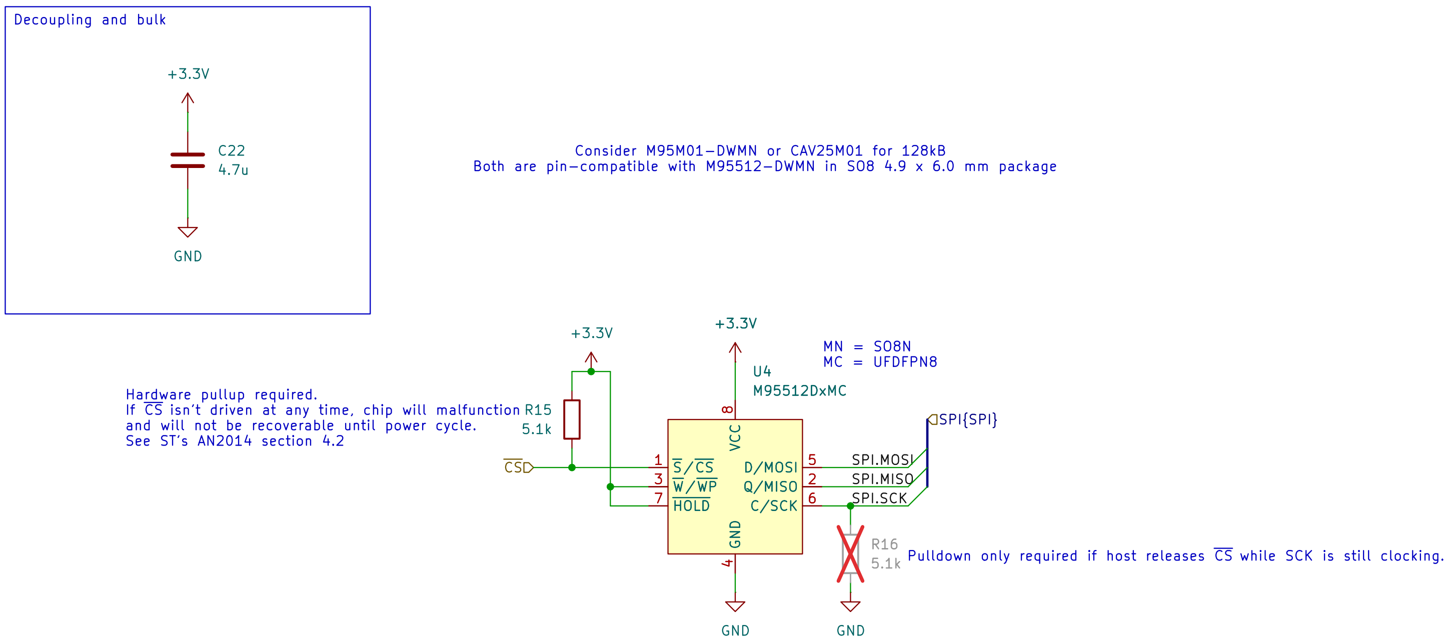




Sheet: /accel/
File: accel_bma400.kicad_sch

Title:

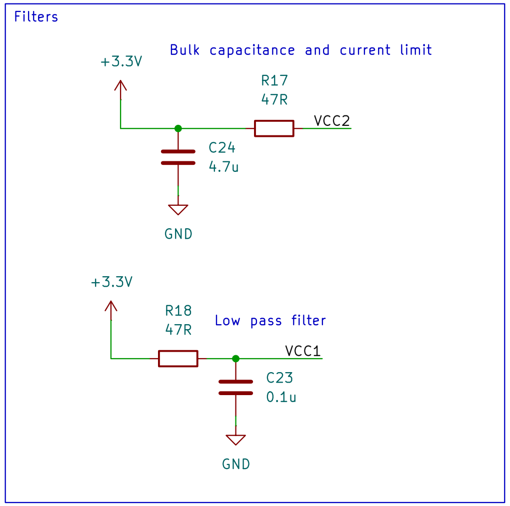
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 6/12



Sheet: /eeprom/
File: eeprom_m95512.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 7/12



See Vishay's app note on reference layouts
https://www.vishay.com/docs/82610/referencelayoutscircuitdiagrams.pdf

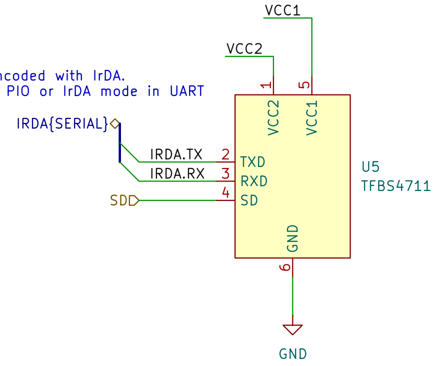
R17 is a current limiting resistor.

47R resistor gives IRED forward current of around 40–50 mA at 3.3 V.
This is 132–150 mW, more than the common 1/16 W (62.5 mW) power rating for resistors.
From testing, this seems to be fine since its pulsed.
Intensity is at around 15%.

100R resistor is 20 mA, or 66 mW, and <10% intensity.

Datasheet says to expect <2 mA average current whtn I_{IRED} (IRED forward current)
is 300 mA and a 20% duty cycle.

Expect serial to be encoded with IrDA.
MCU can do this with PIO or IrDA mode in UART



Original uses IrDA SIR V1.0 at 115200 baud, so we do the same.

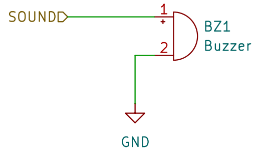
From app note under "emitter signal timing"

"since the receiver saturates during transmission, it is also necessary to include a latency time of > 150 µs after transmission
before the receiver is ready for reception again"

Sheet: /irda/
File: irda_tfdu4711_pio.kicad_sch

Title:

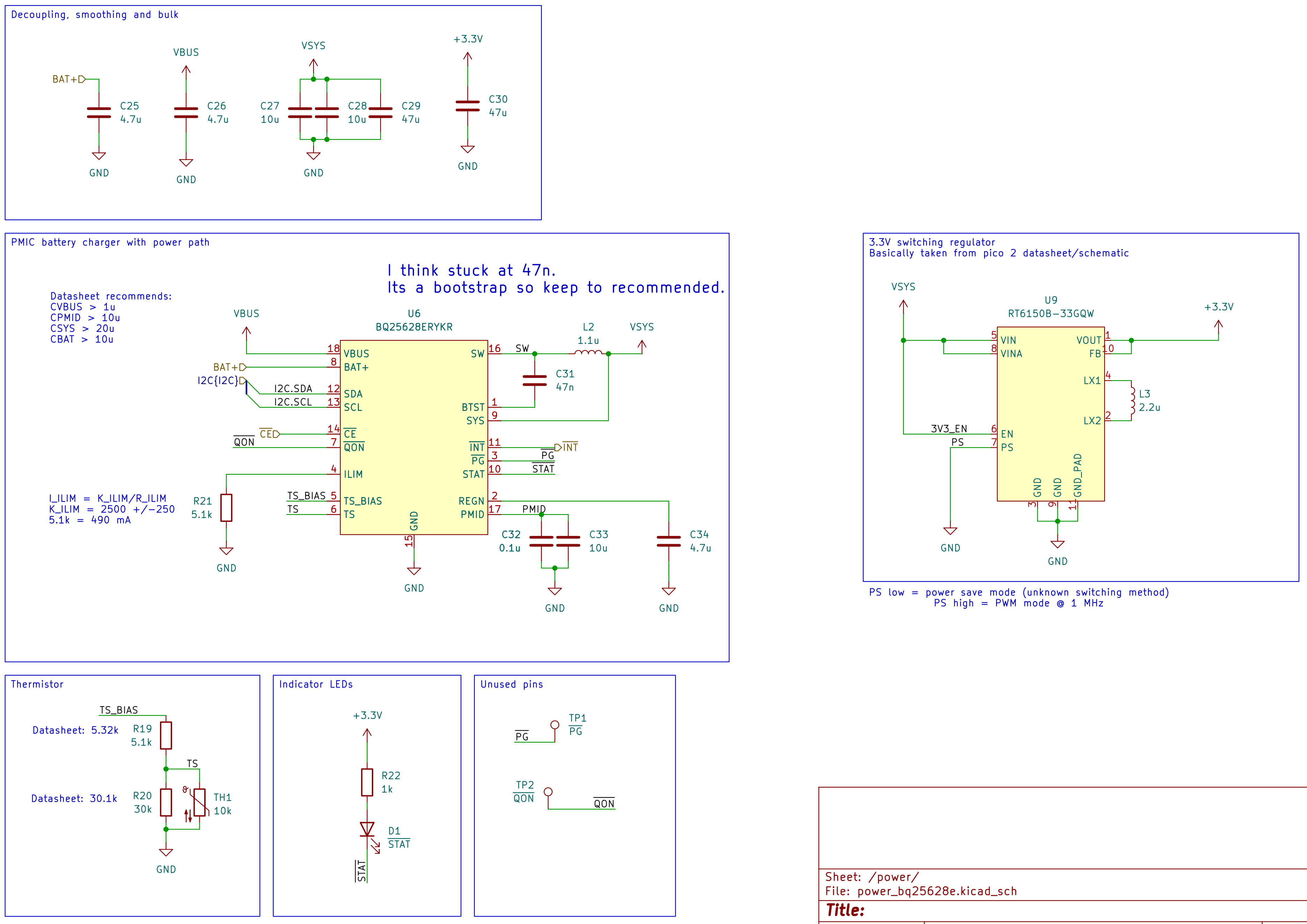
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 8/12



Sheet: /sound/
File: sound_pwm.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 9/12

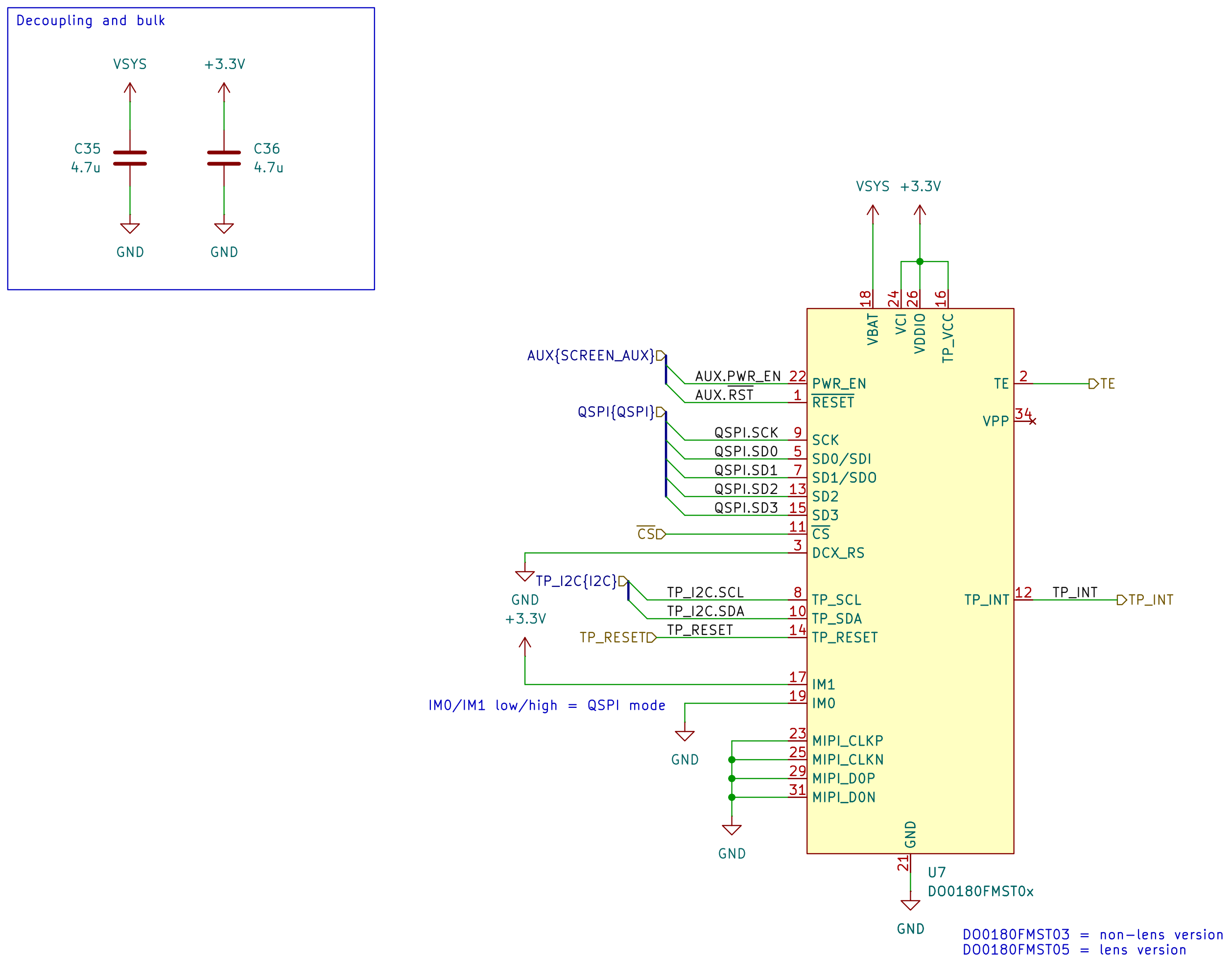


Sheet: /power/
File: power_bq25628e.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.3 | Id: 10/12



Sheet: /screen/
File: screen_do0180fmst03.kicad_sch

Title:

Size: A4 Date:

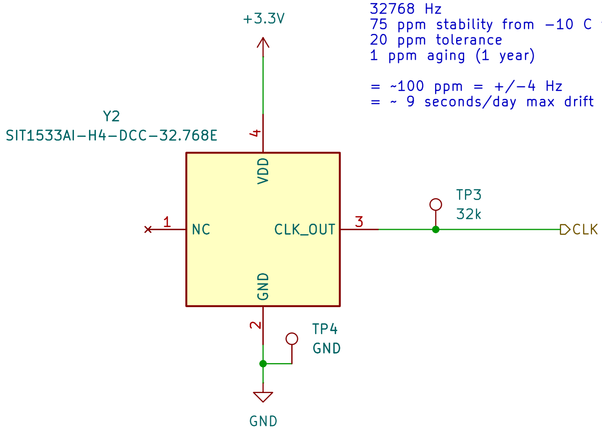
KiCad E.D.A. 10.0.3

Rev:

Id: 11/12

32768 Hz
75 ppm stability from −10 C to +70 C
20 ppm tolerance
1 ppm aging (1 year)

= ~100 ppm = +/−4 Hz
= ~ 9 seconds/day max drift



Sheet: /rtc/
File: rtc_sit1533.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 12/12